📘 AI/ML Internship – Day 67 Notes & Tasks

🟢 Module 9: Introduction to Retrieval-Augmented Generation (RAG) – Building a PDF Question Answering System

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📌 Part 1: Recap of Day 66

Yesterday we learned:

✅ What is RAG?

✅ Knowledge Base

✅ Embeddings

✅ Vector Database

✅ RAG Architecture

Today we will learn:

- How documents are prepared
- Text Chunking
- Embedding Generation
- Vector Database
- Complete PDF Question Answering Pipeline

📌 Part 2: Problem Without RAG

Suppose a company has:

- Employee Handbook (250 Pages)
- AI Internship Guide
- HR Policy
- Cyber Security Manual

User asks:

What is the internship duration?

Without RAG:

❌ AI may not know because these are private documents.

With RAG:

✅ AI searches the document and answers correctly.

📌 Part 3: RAG Pipeline

Upload PDF

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Extract Text

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▼

Split into Chunks

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Create Embeddings

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Store in Vector Database

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User Question

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Convert Question to Embedding

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Similarity Search

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▼

Retrieve Best Chunks

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▼

LLM Generates Final Answer

📌 Part 4: Step 1 – Reading the PDF

What is PDF Text Extraction?

Before AI can understand a PDF, it must extract the text.

Popular Python libraries:

- PyPDF2
- pdfplumber
- PyMuPDF (fitz)

Example

```
from PyPDF2 import PdfReader


reader = PdfReader("internship.pdf")


text = ""


for page in reader.pages:
    text += page.extract_text()


print(text)
```

📌 Part 5: Why Split the Document?

Large PDFs cannot be sent directly to an AI model.

Example:

300-page document

↓

Split into smaller sections.

Example:

Chunk 1

AI Internship Duration

Chunk 2

Eligibility

Chunk 3

Fees

Chunk 4

Projects

📌 Part 6: What is Chunking?

Definition

Chunking is the process of dividing a large document into smaller pieces.

Example

Original Paragraph

Artificial Intelligence is changing every industry...

↓

Split into:

Chunk 1

Chunk 2

Chunk 3

Each chunk contains approximately 300–500 words.

Why Chunking?

✅ Faster searching

✅ Better accuracy

✅ Lower API cost

📌 Part 7: Creating Embeddings

After chunking,

each chunk becomes a numerical vector.

Example:

Chunk

Course Duration is 3 Months

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Embedding

[0.42, 0.81, 0.23, ...]

Computers compare these vectors to find similar meanings.

📌 Part 8: Storing in Vector Database

The embeddings are stored in a Vector Database.

Popular Vector Databases:

- FAISS
- ChromaDB
- Pinecone
- Weaviate

Example:

Chunk	Vector
Fees	0.81,0.33...
Duration	0.72,0.44...
Hostel	0.18,0.61...

📌 Part 9: User Asks a Question

User:

How long is the internship?

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Question is converted into an embedding.

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Vector Database searches for the most similar chunk.

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Retrieved Chunk:

Internship Duration: 3 Months

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AI generates:

The internship duration is 3 months.

📌 Part 10: Similarity Search

The vector database compares:

Question Embedding

↓

Document Embeddings

↓

Most similar document is selected.

This is called **Similarity Search**.

📌 Part 11: Complete RAG Architecture

```
         User
          |
          ▼
```

```
   Ask Question
        |
        ▼
 Convert to Embedding
        |
        ▼
   Vector Database
        |
   Similarity Search
        |
        ▼
Retrieve Relevant Chunks
        |
        ▼
   Large Language Model
        |
        ▼
    Generate Answer
```

📌 Part 12: Real-World Applications

Company Chatbot

Reads HR documents.

Hospital Assistant

Answers from medical guidelines.

College Assistant

Reads admission brochures.

Legal Assistant

Reads contracts.

Banking Assistant

Searches banking regulations.

Research Assistant

Answers questions from research papers.

📌 Part 13: Advantages

✅ Uses company documents

✅ Uses latest information

✅ High accuracy

✅ Less hallucination

✅ Easy to update (just add new documents)

📌 Part 14: Limitations

❌ Needs clean documents

❌ Requires embeddings

❌ Retrieval quality affects answers

❌ Slightly slower than answering directly

📌 Part 15: Theory Questions

What is Chunking?

Chunking is the process of dividing a large document into smaller pieces for efficient searching and processing.

Why is chunking required in RAG?

Large documents cannot be processed efficiently in one request. Chunking improves retrieval speed and accuracy.

What is Similarity Search?

Similarity Search finds the document chunks that are most semantically similar to the user's question.

Why are embeddings created?

Embeddings convert text into numerical vectors so computers can compare meanings instead of exact words.

What is the purpose of a Vector Database?

A vector database stores embeddings and retrieves the most relevant information using similarity search.

📌 **Part 16: Interview Questions**

Explain the RAG pipeline.

A RAG pipeline extracts text from documents, splits it into chunks, converts the chunks into embeddings, stores them in a vector database, retrieves relevant chunks based on the user's query, and finally uses an LLM to generate an answer.

Why do we split documents into chunks?

To improve retrieval accuracy, reduce processing cost, and handle large documents efficiently.

What is an embedding?

An embedding is a numerical representation of text that captures its semantic meaning.

Name three vector databases.

- FAISS
- ChromaDB
- Pinecone

What is similarity search?

It is the process of finding the most relevant document chunks by comparing vector embeddings.

🎯 Tasks for Day 67

🧠 Theory

1. Explain the complete RAG pipeline.
2. Define chunking.
3. What is similarity search?
4. Why are embeddings important?
5. Explain the role of a vector database.

💻 Practical Task 1

Install the required libraries:

pip install PyPDF2

pip install chromadb

pip install sentence-transformers

💻 Practical Task 2

Read a PDF file using **PyPDF2** and print its contents.

💻 Practical Task 3

Create a sample document and manually divide it into **5 chunks**.

💻 **Practical Task 4**

Draw the complete RAG pipeline diagram.

💻 **Practical Task 5**

Research the differences between:

- FAISS
- ChromaDB
- Pinecone

Prepare a comparison table.